

ERIK NORRIS

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Sr. Staff Mechanical Engineer / Designer | Forensic Architect

// AUTODIDACT TECHNICAL POLYGLOT | CONCEPT TO PRODUCTION | BBQS TO DENTAL CHAIRS

PROFESSIONAL SUMMARY

I architect the hardware interface for the physical world—from the heat of a Fissler BBQ to the surgical precision of a KaVo Dental Chair. My career is defined by a relentless drive to translate 'Concept' into 'Mass Production' across every conceivable domain. I have helped validate the most beautiful PC ever made (SGI Indigo), engineered the Gold Standard for pro audio consoles (Avid D-Control), and optimized the OG gaming platform (Xbox). Whether it's managing the 150-watt thermal load of a retinal headset, ensuring IP69K washdown compliance for food robotics, or guiding a Class III cardiac catheter through the FDA—the physics change, but the mission remains the same: Zero Yield Loss. Zero Cosmetic Defects. Total Functional Integrity.

CORE COMPETENCIES

ENGINEERING

- Forensic Engineering
- Mechanical Architecture
- Complex Surface Modeling (Pro/E, Creo)
- Mechanism Design
- Thermal Management (Active/Passive)

MANUFACTURING

- New Product Introduction (NPI)
- Design for Manufacturing (DFM)
- Root Cause Analysis & Yield Improvement
- Injection Molding (Class A)
- Die Casting & Thixomolding
- Sheet Metal (Progressive)

TOOLS & REGULATORY

- PTC Creo / Pro/ENGINEER
- PLM (Windchill Admin)
- Onshape / SolidWorks
- GD&T / Tolerance Analysis
- Technical Program Management
- Supplier Qualification & Strategy
- Regulatory Compliance (UL/FCC/CE)
- Class III Medical Device Standards

PROFESSIONAL EXPERIENCE

HYPHEN | Mechanical Engineer

2021 - 2022

| Lead engineer for the 'Factory-in-a-Box' food robotics platform.

- Integrated 350+ electromechanical actuators into a constrained chassis.
- Achieved **IP69K compliance** for wash-down environments, designing custom sealing solutions for doors, cable pass-throughs, and sensor arrays.
- Reduced **Mean Time To Repair (MTTR)** from hours to minutes by architecting a hot-swappable module strategy that decoupled dispensers from the main chassis.

NOON HOME | Senior Mechanical Engineer

2017 - 2018

| Lead mechanical architect for the 'Haptic Switch' smart lighting controller.

- Designed a floating flexure system to isolate the OLED display stack from mounting forces, resulting in **zero cosmetic yield loss** at assembly.
- Collaborated with ID to achieve 'dead-front' aesthetics using precision tint-matching and seamless glass integration.

AVEGANT | Senior Mechanical Engineer

2015 - 2017

Mechanical Lead for mixed-reality retinal imaging headsets.

- **Thermal Engineering:** Utilized the magnesium headband as a passive heat sink to dissipate 150W equivalent heat load, resolving a critical 'Hot Head' user complaint.
- **Mechanism Design:** Engineered a high-precision IPD adjustment mechanism, ensuring optical alignment stability under drop-test conditions.

KALEIDESCAPE | Senior Mechanical Design Engineer

2008 - 2015

Sole Mechanical Engineer responsible for the M500, M300, and Cinema One product lines.

- **Crisis Management:** Detected 'Flow Mark' defects in 1,200 parts (Cinema One). Exercised **Line-Down** authority to reject inventory and force a root-cause gating change, establishing 'Apple-tier' cosmetic standards.
- **Cost Reduction:** Redesigned the 'Macduff' chassis lid, consolidating three parts into one. Reduced assembly time by **15 minutes** and eliminated fastener count by 30%.
- **System Integration:** Integrated slot-loading optical drives (Blu-ray) with custom vibration isolation dampers to prevent skip/read errors during operation.

DIGIDESIGN (AVID TECHNOLOGY) | Lead Mechanical Engineer / Industrial Designer

2003 - 2008

Led flagship console architecture (D-Control, C|24, SC48).

- **SC48 'Lux' Console:** Solved a critical 75°C CPU thermal shutdown during DVT by redesigning the chassis height from 3U to 4U and optimizing internal airflow paths. Eliminated custom aluminum extrusions, reducing chassis cost by **20%**.
- **C|24 Governance:** Enforced strict **Data Control Drawing (DCD)** protocols for the integration of 19 PCBs, achieving 100% mechanical fit on the first pilot build.
- **C|24 Manufacturing:** Designed a 'Vertical Hanging' fixture for painting ABS sidecaps, solving a 2mm warp/shrinkage issue that was causing assembly failure.

WEBTV / MICROSOFT | Mechanical Engineering Consultant

1999 - 2003

High-level surfacing and design consultancy via Mechanistic.

- **Project Cortez (Wireless Keyboard):** Delivered complex, organic Pro/E surfaces for the infrared keyboard in **6 weeks**. Resolved a critical 0.05mm key pitch mismatch between ID intent and Silitek's tooling.
- **Project Elmer/Zeus (Servers):** Engineered rapid-prototype server enclosures for CES. Managed a hybrid build (soft-tool metal + CNC plastic) to deliver 5 functional units in **4 weeks**.
- **Xbox Division:** Transitioned engineering assets from UltimateTV to Xbox, designing EMI liners and HDD carriers for early development kits.

MECHANISTIC | Principal / Freelance Product Designer

1999 - 2003

Provided expert CAD surfacing and mechanical engineering services.

- Provided expert CAD surfacing and mechanical engineering services for clients including Apple, Motorola, and SGI.
- Designed complex organic surfaces for consumer peripherals using Pro/ENGINEER.

FROGDESIGN | Mechanical Designer

1997 - 1998

Bridged the gap between high-concept Industrial Design and mass production.

- **Design Fidelity:** Maintained subtle surface, gap, and finish requirements while ensuring manufacturability for clients like KaVo (Dental Systems) and Newscorp (Satellite Receivers).

Roles establishing the foundation of high-performance product design.

- **Mechanistic (1993–1996):** Designed SwitchBlade inline speed skates and Sunbeam Toast Logic appliances.
- **Silicon Graphics (SGI) (1989–1993):** Performed thermal/acoustic analysis for Indigo and Indy workstations; managed prototype supply chains.
- **EP Technologies (1986–1989):** Production Supervisor for Class III medical devices (Cardiac Ablation Catheters).

EDUCATION

De Anza College
Associate’s Degree, Automotive Engine Performance

AWARDS & PATENTS

- Avegant Glyph (Best of CES 2016)
- Kaleidescape Cinema One (CEPro Product of the Year)
- Motorola MP3 Player (CES Honoree) - Mechanical Design & Surfacing